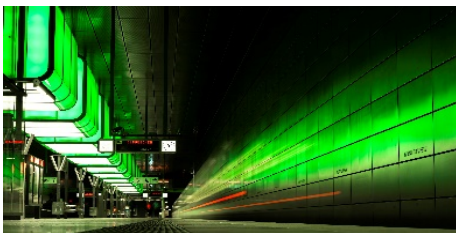


LED & Signage/Display Applications

LED light Applications in Railway Stations



Source: Google Keyword Image Search

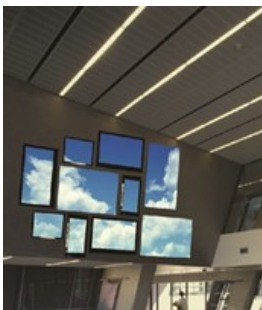
Signage and LED Display Applications in Railway Stations



3

My related Achievements

Smart Flexible Signage: Development of Wireless HDMI Interface and Applications



Smart flexible signage applications developed using a wireless HDMI (High-Definition Multimedia Interface) module with KETI and a partner company

My related Achievements

Smart flexible signage applications with Wireless HDMI



These technologies can be usefully used in exhibition and information displays in railway stations

My related Achievements

Transparent Display based AR(Augmented Reality)



This kind of AR technology can be used in window-style displays in railway stations

My related Achievements

Transparent & multi-layered 3D Display Applications



- 49 -

My related Achievements

Main topics of IoL(Light IoT) Research Center (ITRC)

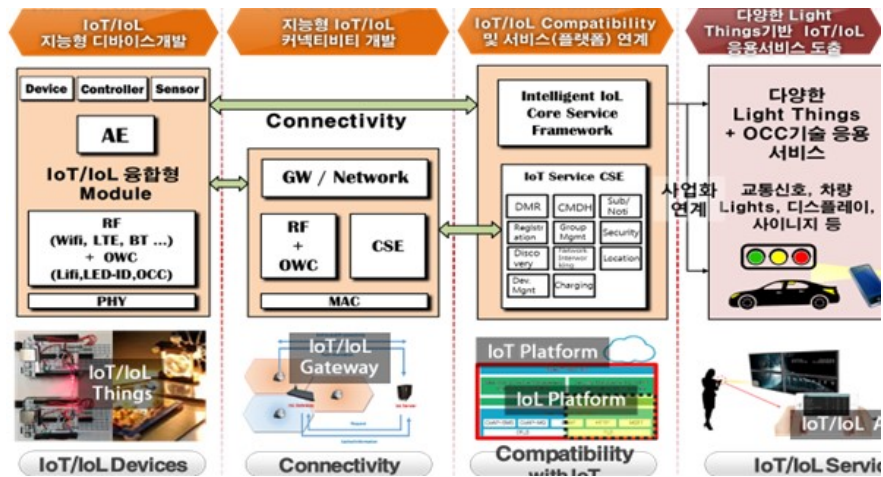


Above topics of ITRC center are closely related with today,s presentation topics

- 52 -

My related Achievements

Main topics of IoL(Light IoT) Research Center (ITRC)



Above topics of ITRC center are closely related with today,s presentation topics

- 53 -

My related Achievements

IoL applications & Communication method



source: IEEE 802 document

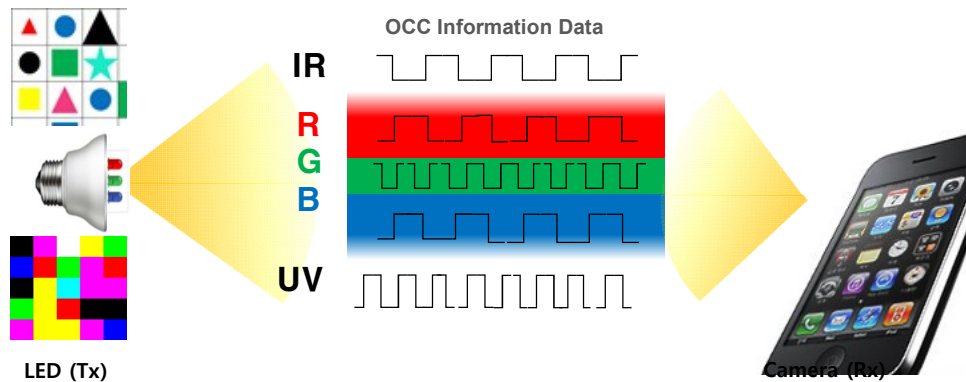
LiFi & Optical Camera Communication

(LiFi = Ligth WiFi concept), OCC = Optical Camera Communication

- 54 -

My related Achievements

IEEE802.15.7m Standard technology : OCC



Tx – LED light source, Rx – camera (image sensor)

In this system , Camera could be a receiver (Simple Rx implementation)

My related Achievements

OCC application case



Tx : Display Panel Rx : Smart Device Camera

My related Achievements

OCC application case

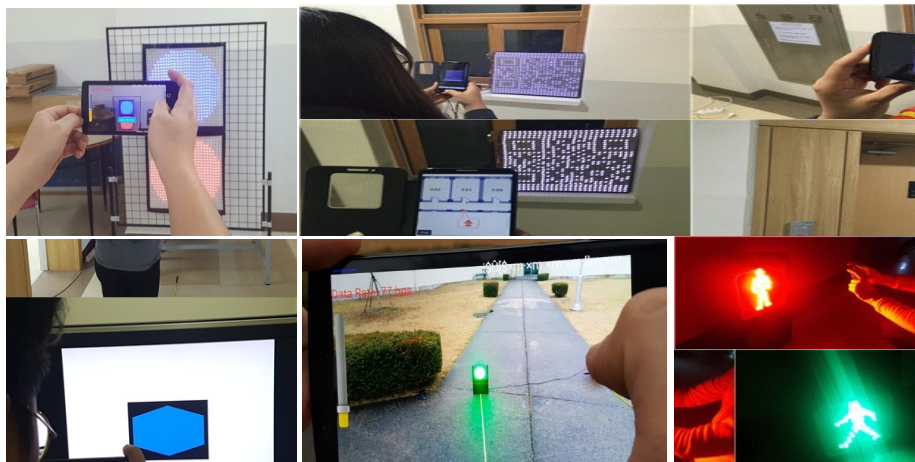


Tx : LED color tag ID Rx : Smart Device Camera

- 57 -

My related Achievements

OCC application



Smartphone camera-based OCC data recovery test was conducted.

- 58 -

My related Achievements

OCC application

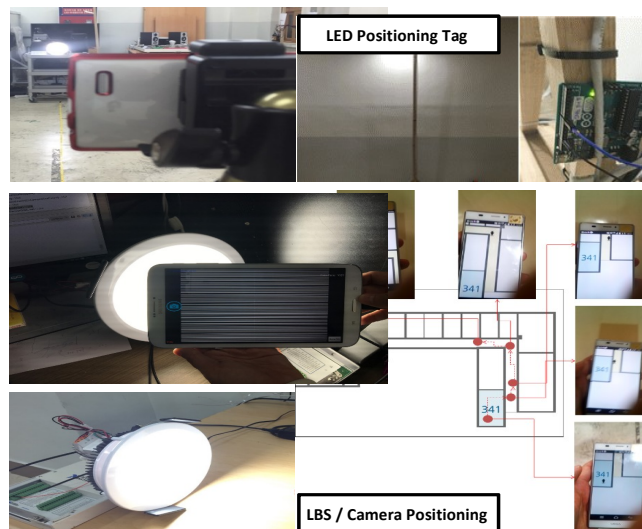


Tx : LED tag (Display) or Light Rx : Smart Device Camera

- 59 -

My related Achievements

OCC application



Location based (positioning) service is also available

- 60 -

My related Achievements

OCC application



Location based (positioning) service is also available

My related Achievements

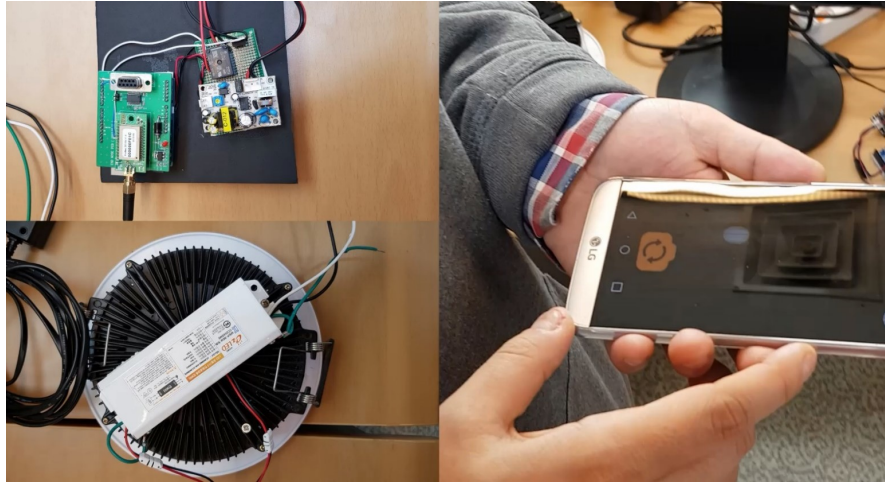
LiFi application



Tx – LED light source, Rx – photo Diode

My related Achievements

Wireless solutions for LED light control



Wi-Fi/BT-connected LED lights are controlled by smart devices

- 63 -

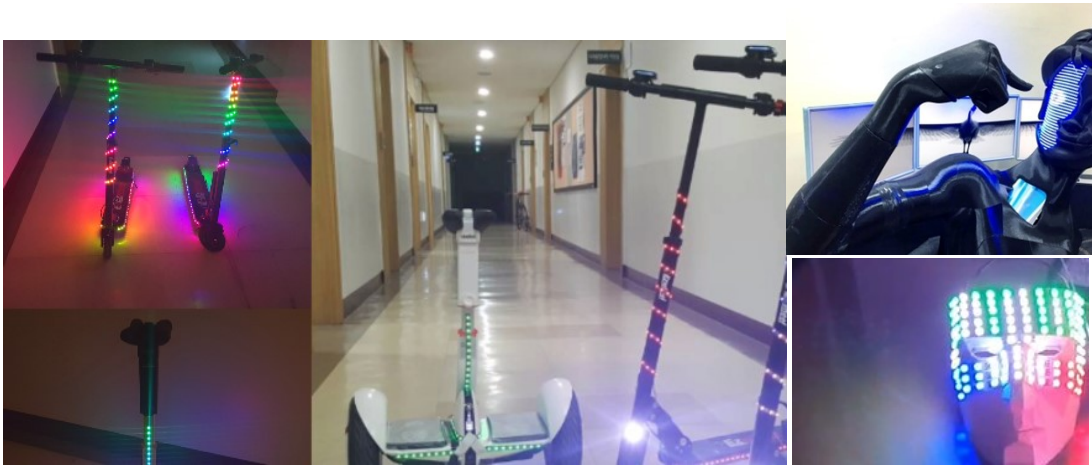
LED light On/off control by smart device



- 64 -

My related Achievements

Attachable LED applications



- 65 -

My related Achievements

Attachable LED for Ride applications

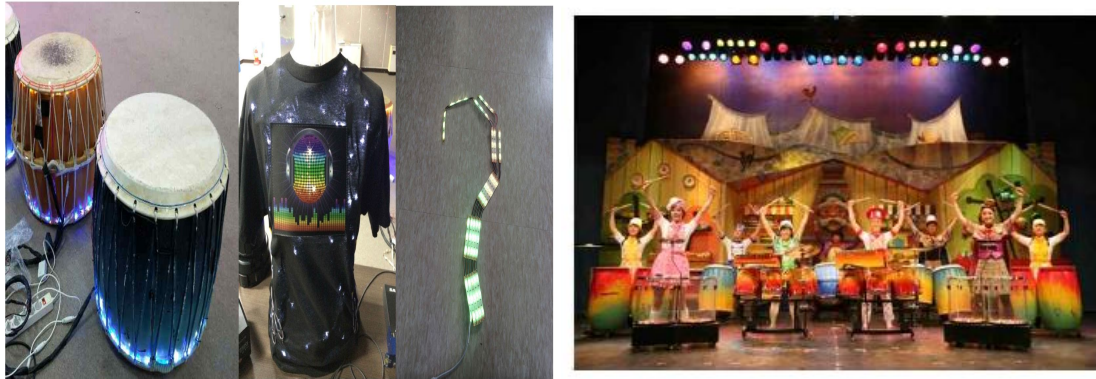


Electric Generator connected LED patch for BIKE

- 66 -

My related Achievements

Attachable LED applications

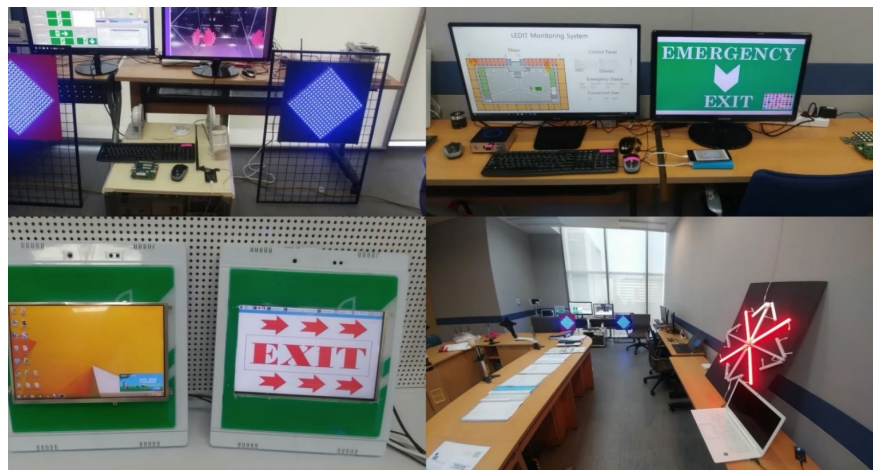


This kind of LED patch was used in the “Nanta” performance by PMC. This attachable LED patch is very useful for the **safety of railway track workers**.

- 67 -

My related Achievements

Guide LED indicating EXIT direction

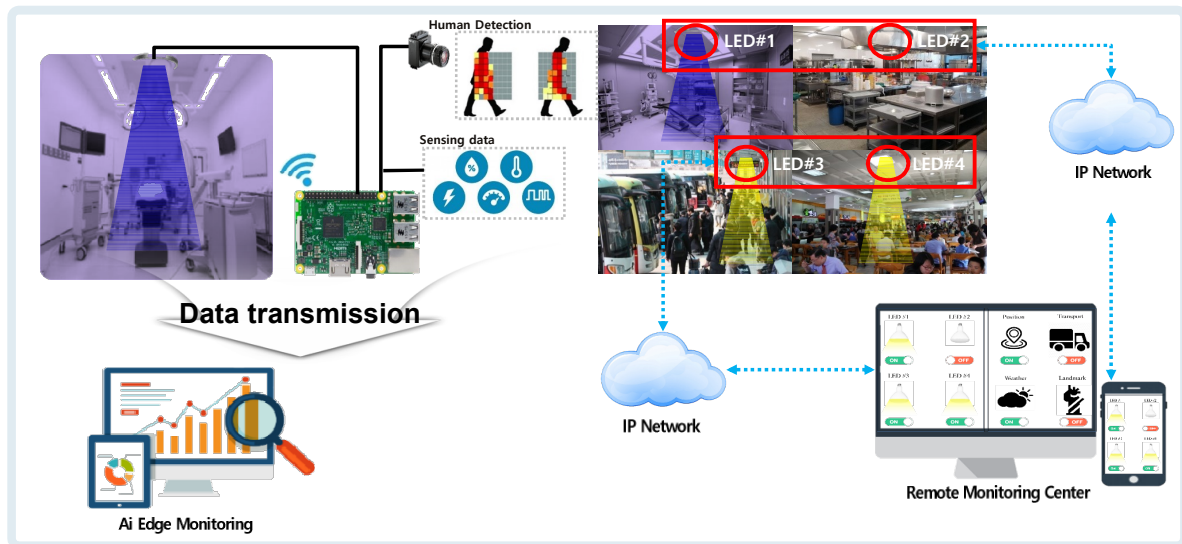


This EXIT guide LED can be used effectively in railway stations

- 68 -

My related Achievements

Thermal Imaging Camera based Human Detection + Sanitary light control + Monitoring



Thank you for listening

Appendix 1

(Previous Standard activities : ieee802)

Jaesang Cha Chiba Univ., Japan

- 93 -

Activity Introduction

◆ Internationals standard activitis

IEEE 802.15.7m SG (2012 ~ 2013)
IEEE 802.15.7m SG (2014)
IEEE 802.15.7m SG (2016.11 ~ 현재)
802.15.7m IG VAI(CamGo)
802.15.7m IG (180, LIFU)
802.11.5m IG (LG-TIG)

IEEE 802.15.7m TG(SG) standard docu.

ITU-R SG1 standard proposals

IEEE 802.11/15. standard meeting

ITU-R SG1 standard meeting

- 94 -